

Day 4: Power Query Fundamentals & Query Editor

📌 Definition to Remember

Power Query is the ETL (Extract, Transform, Load) data preparation engine in Power BI used to clean, reshape, and transform raw data before loading it into the data model.

1. Power Query Editor Interface

- **Queries Pane (Left):** Lists all data connections and staging queries.
- **Data Preview Grid (Center):** Interactive tabular preview of transformations.
- **Applied Steps Pane (Right):** Sequential recorded pipeline of transformations executed from top to bottom.
- **Formula Bar (Top):** Displays the underlying **M Language code** for the selected transformation step.

2. Fundamental Transformation Operations

1. **Promote First Row as Headers:** Converts top data row into official column names.
2. **Explicit Data Typing:** Setting accurate types (`Int64` , `Decimal` , `Text` , `Date` , `DateTime` , `True/False`) to avoid memory bloat and calculation bugs.
3. **Remove / Keep Columns:** Eliminating unnecessary columns at the ETL stage to minimize semantic model size.
4. **Row Filtering & Sorting:** Filtering out irrelevant historical years, test records, or blank rows.
5. **Rename & Reorder Columns:** Creating clean, business-friendly attribute names for end-user clarity.

3. Query Properties

- **Enable Load:** Unchecking prevents staging/temporary lookup queries from consuming RAM in the report model.
- **Include in Report Refresh:** Disabling static tables (e.g., country codes) to accelerate daily refresh times.

★ Must-Master Skills & Interview Points

1. All Power Query steps are non-destructive and replayable via the **Applied Steps** chain.
2. Always filter and remove unused columns as early as possible in Power Query.
3. Understand why explicit data typing prevents silent DAX type coercion errors.

⚡ Quick Recall

Extract → Promote Headers → Enforce Data Types → Filter & Drop Columns → Applied Steps Pipeline → Load